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# Pd-Catalyzed Regio- and Stereoselective Cyclization–Heck Reaction of Monoesters of 1,2-Allenyl Phosphonic Acids with Alkenes

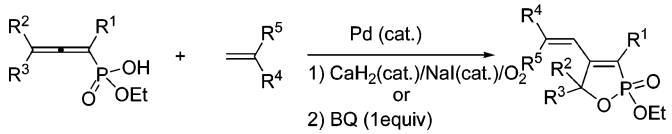
**Fei Yu, Xiongdong Lian, and Shengming Ma***

*State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 354 Fenglin Lu, Shanghai 200032, People's Republic of China*

*masm@mail.sioc.ac.cn*

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## ABSTRACT




**The cyclization–Heck reactions of monoesters of 1, 2-allenyl phosphonic acids with alkenes were studied. The reaction afforded 4-(1-*Z*-alkenyl)-2-ethoxy-2,5-dihydro[1,2]oxaphosphole 2-oxides regio- and stereoselectively. Pd(II) was regenerated from the in situ formed Pd(0) using $CaH_2$(cat.)/NaI/$O_2$ or benzoquinone to furnish the catalytic cycle.**

In transition-metal-catalyzed organic transformations, palladium-catalyzed reactions have been emerging as the most powerful methods for the formation of carbon−carbon and carbon−heteroatom bonds.[1] Among these reactions, the Heck reaction has been intensively studied[2,3] and applied to the total synthesis of many natural products[4] because of its high versatility. Although the coupling−cyclization of functionalized allenes[5,6] with aryl/alkenyl halide,[7,8] allylic halide,[9] ($\omega$-1)-alkenyl halide,[10] propargylic carbonates,[11] or 2,3-allenols[9d,12] has been demonstrated, the coupling−cyclization of these allenes with alkenes, which may provide an efficient pathway to the alkenyl-substituted cyclic compounds, has not been studied. A palladium-catalyzed mechanism may be envisioned for such a reaction: *endo*-mode cyclic nucleopalladation of allenes **I** would form cyclic palladium intermediates **II**. Then, the carbon−carbon double bond in the alkenes

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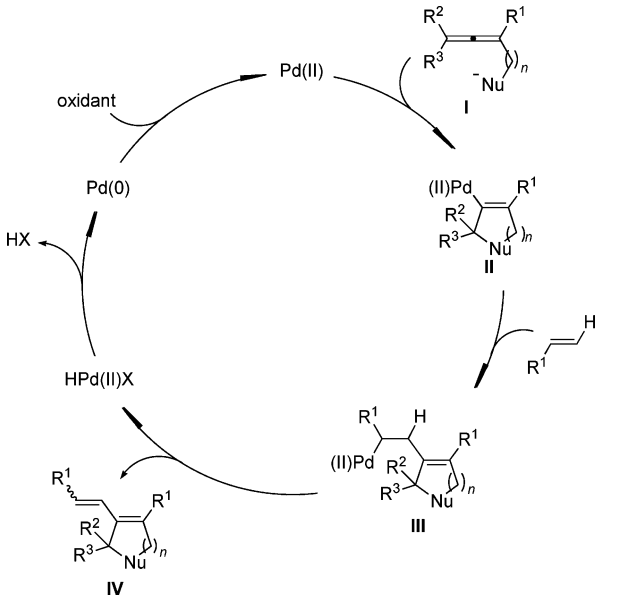
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inserts into the C−Pd bond in intermediates **II** to form intermediates **III**, which would afford the products **IV** via $\beta$-H elimination. The in situ generated palladium hydride species would form Pd(0) species.[13] To make the reaction catalytic in Pd, the Pd(0) species must be oxidized to the catalytically active Pd(II) species (Scheme 1). On the other hand, many heterocycles that include a phosphorus atom and an oxygen atom usually have important biological activities.[14] Although some methods have been developed,[15] development of a highly efficient method for the synthesis of these molecules with diversity is still of high interest. In this paper, we disclose our recent observation on the cyclization−Heck reaction of monoesters of 1,2-allenyl phosphonic acids with alkenes.

**Scheme 1.** Proposed Catalytic Cycle for Palladium-Mediated Cyclization−Heck Reaction




Under the catalysis of $PdCl_2(PhCN)_2$, only 14% yield of cyclization product **2a** was formed together with a significant amount of chlorocyclization product **3a** when $CuCl_2$ (4.0 equiv) and $K_2CO_3$ (4.0 equiv) were used (Scheme 2).[16] When the reaction was conducted in air with a $CaCl_2$ drying tube in the presence of 0.2 equiv of NaI,[17] *E*-**2a** was formed stereoselectively (*E*/*Z* ratio = 7:1) in 49% yield (entry 1, Table 1). To our surprise, when a catalytic amount of $CaH_2$ was added, the yield of **2a** increased significantly (entry 2, Table 1). However, when we increased the amount of $CaH_2$ further, lower yields and the formation of palladium black were observed (entries 3 and 4, Table 1). A catalytic amount of NaH gave a very similar result under comparable

**Scheme 2**

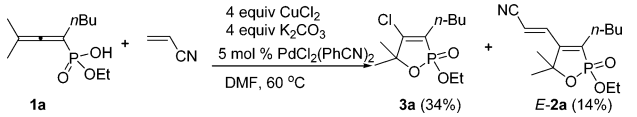
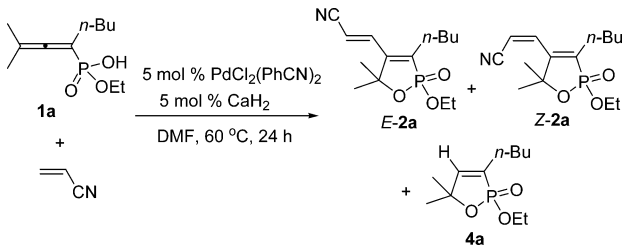



**Table 1.** Pd-Catalyzed Cyclization−Heck Reaction of Ethyl (2-Methylocta-2,3-dien-4-yl)phosphonic Acid **1a** with Acrylonitrile[a]




| entry | oxidant (0.2 equiv) | yield of *E*-**2a** (%)[b] | yield of *Z*-**2a** (%)[b] |
|---|---|---|---|
| 1[c] | NaI | 49 | 7 |
| 2 | NaI | 70 | 4 |
| 3[c] | NaI | 16 | 2 |
| 4[d] | NaI | 9 | trace |
| 5[e] | NaI | 70 | 7 |
| 6[d] | NaI | 41 | 10 |
| 7[f] | NaI | 9 | 1 |
| 8 | − | 27 | 7 |
| 9 | BQ[g] | 35 | 6 |
| 10 | $CuCl_2$ | 38 | 17 |
| 11 | KI | 34 | 4 |
| 12 | *n*-BuI | 46 | 4 |
| 13 | NaBr | 34 | 6 |
| 14 | NaCl | 9 | 2 |

[a] The reactions were conducted in air with a $CaCl_2$ drying tube unless otherwise stated. [b] Determined by $^1H$ NMR analysis with $CH_2Br_2$ as the internal standard. [c] With 0.5 equiv of $CaH_2$. [d] With 1 equiv of $CaH_2$. [e] 0.05 equiv of NaH in air without a $CaCl_2$ drying tube. [f] In an Ar atmosphere. [g] Benzoquinone.


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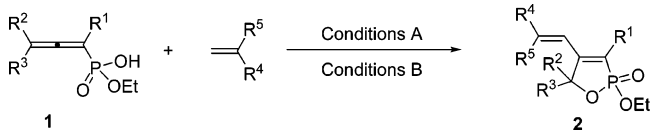
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**Table 2.** Cyclization−Heck Reaction of Monoesters of 1,2-Allenyl Phosphonic Acids **1** with Alkenes[a]




| | **1** | | | | alkenes | | | | |
|---|---|---|---|---|---|---|---|---|---|
| entry | $R^1$ | $R^2$ | $R^3$ | | $R^4$ | $R^5$ | conditions | yield of *E*-**2** (%)[b] | *E*/*Z*[c] |
| 1 | *n*-Bu | Me | Me | (**1a**) | COOMe | H | A | 88 (**2b**) | >99:1 |
| 2 | *n*-Bu | Me | Me | (**1a**) | COOEt | H | A | 84 (**2c**) | >99:1 |
| 3[d] | *n*-Bu | Me | Me | (**1a**) | Ph | H | A | 52 (**2d**)[e] | [f] |
| 4 | *n*-Bu | Me | Me | (**1a**) | Ph | H | B | 85 (**2d**) | [f] |
| 5 | *n*-Bu | $-(CH_2)_4-$ | | (**1e**) | COOMe | H | A | 69 (**2e**) | >99:1 |
| 6 | Me | $-(CH_2)_5-$ | | (**1f**) | COOMe | H | A | 79 (**2f**) | >99:1 |
| 7 | Bn | Me | Me | (**1g**) | COOMe | H | A | 75 (**2g**) | [f] |
| 8 | *n*-$C_6H_{13}$ | Et | Et | (**1h**) | COOMe | H | A | 85 (**2h**) | >99:1 |
| 9 | *n*-Bu | Me | Me | (**1a**) | COOMe | Me | B | 61 (**2i**) | >99:1 |
| 10 | *n*-Bu | Me | Me | (**1a**) | $COONH_2$ | H | B | 86 (**2j**) | >99:1 |
| 11 | *n*-Hept | H | Me | (**1k**) | COOEt | H | B | 63 (**2k**) | 55:1 |
| 12 | *n*-Hept | H | Me | (**1k**) | Ph | H | B | 63 (**2l**) | [f] |
| 13 | *n*-Bu | H | H | (**1m**) | COOEt | H | B | 46 (**2m**) | >99:1 |
| 14 | *n*-Bu | H | H | (**1m**) | $COONH_2$ | H | B | 48 (**2n**) | [f] |
| 15 | H | $-(CH_2)_5-$ | | (**1o**) | COOEt | H | B | 61 (**2o**) | >99:1 |
| 16 | H | $-(CH_2)_5-$ | | (**1o**) | $COONH_2$ | H | B | 34 (**2p**) | [f] |

[a] Conditions A: 1.0 equiv of **1**, 5.0 equiv of alkene, 0.2 equiv of NaI, 5 mol % of $PdCl_2(PhCN)_2$, and 5 mol % of $CaH_2$ in DMF in air with a drying tube at 60 °C for 24 h. Conditions B: 1.0 equiv of **1**, 5.0 equiv of alkene, 5 mol % of $PdCl_2(PhCN)_2$, and 1.0 equiv of BQ in DMF at 60 °C overnight. [b] Isolated yields. [c] Determined by $^1H$ NMR spectra of the crude products. [d] The reaction was conducted at 100 °C. [e] The cycloisomerization product **4a** (25%) was isolated. [f] Only the signal of the *E*-isomers can be identified in the $^1H$ NMR spectra.

conditions (entry 5, Table 1). It is reasoned that a catalytic amount of $CaH_2$ or NaH may help the oxypalladation step in the beginning. However, in the presence of 0.5 or 1 equiv of $CaH_2$, the generation of Pd(0) from HPd(II)X may be so fast that the NaI−$O_2$ system[17] may not be able to oxidize it back to the catalytically active species Pd(II) efficiently enough, thus forming Pd black. When the reaction was conducted in air without a drying tube in Ar or $N_2$ atmosphere, or in the absence of NaI, the yields of **2a** decreased significantly (entries 6−8, Table 1). The yields of **2a** also dropped with a catalytic amount of benzoquinone (BQ) and some other organic or inorganic halides (entries 9−14, Table 1). In all these reactions, the formation of **4a** was less than 2%, if any.

With the established standard conditions in hand, we studied the scope of the reaction (Table 2). The cyclization−Heck reaction products were highly regio- and stereoselectively formed in high yields as a single isomer when acrylic acid esters were used (entries 1 and 2, Table 2). Some other fully substituted monoesters of 1,2-allenyl phosphonic acids with methyl acrylate also smoothly afforded the expected products in high yields and high *E*/*Z* ratios (entries 5−8, Table 2). However, this procedure is limited for some less reactive alkenes such as styrene, acrylamide or 2-methylpropenoic acid methyl ester, and 3-monosubstituted or 3-unsubstituted allenes. For example, we observed that when 1,2-allenyl phosphonic acid **1a** was reacted with styrene the desired product **2d** was obtained initially in 52% yield together with byproduct **4a** in 25% yield (entry 3, Table 2). However, this complication could be easily overcome by using BQ as the oxidant to produce the product **2d** in 85% yield as the only product (entry 4, Table 2). With this new procedure, 2-methylpropenoic acid methyl ester and acrylamide can also react with **1a** to afford **2i** and **2j** in 64% and 66% yields, respectively (entries 9 and 10, Table 2). This procedure can also be extended to the 3-monosubstituted or 3-unsubstituted allenes **1k**, **1n**, and **1o**: their reaction with ethyl acrylate, styrene, or acrylamide afforded the corresponding products **2k**−**p**, respectively (entries 11−16, Table 2).

In conclusion, we have established the cyclization−Heck reaction of 1,2-allenyl phosphonic acids with alkenes under oxidation conditions: $CaH_2$(cat.)/NaI/$O_2$ or benzoquinone. Under the standard conditions, the cyclization−Heck reaction

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products were formed highly regio- and stereoselectively. Further studies in this area including the scope of substrates and the determination of the role of $CaH_2$ are being conducted in our laboratory.

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**Supporting Information Available:** Synthesis and characterization for **2** and **3a**. This material is available free of charge via the Internet at http://pubs.acs.org.

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